

Course Code: CS-1002

Name: M. Amaan Masood

Course Name: Prog Fundamentals Roll No: 26K-2524

Teacher Name: Syed Faraz

Dept: BDS

FOR I TO N Ahmed Zaidi

Section: 1A

Part A - Foundational Practice Problems.

Answers:

Q1) Pseudocode

OUTPUT "Enter a number"

INPUT N // N is an integer

WHILE $N \neq 0$ DO

OUTPUT N

 $N = N - 1$

ENDWHILE

Q2) Pseudocode

INPUT Year

INPUT Year

IF $\text{Year} \bmod 4 = 0$ THEN

OUTPUT "Leap Year"

ELSE

 IF $\text{Year} \bmod 100 \neq 0$ OR $\text{Year} \bmod 400 = 0$ THEN

OUTPUT "Leap Year"

ELSE

OUTPUT "Not Leap Year"

ENDIF

ENDIF

Q3)

Pseudocode

OUTPUT "Enter a number"

INPUT Num

StrNum $\leftarrow$ STR(Num)

Stringlength $\leftarrow$ LENGTH(StrNum)

IF Stringlength = 1 THEN

 OUTPUT "single-digit number"

ELSE

 IF Stringlength = 2 THEN

 OUTPUT "double-digit number"

 ELSE

 IF Stringlength = 3 THEN

 OUTPUT "triple-digit number"

 ELSE

 OUTPUT Num, "digit number"

ENDIF

Q4) Pseudocode!

start $\leftarrow$ 1, count $\leftarrow$ 1

3

WHILE start $\leq$ 10 DO

6

FOR count 1 TO 10

9

OUTPUT start * count

start $\leftarrow$ start + 1

IF start = 3 OR start = 6 OR start = 9 THEN

start $\leftarrow$ start + 1

ENDIF

ENDWHILE

Q5) Pseudocode.

Sum $\leftarrow$ 0

OUTPUT "Enter a number"

INPUT Number

number string $\leftarrow$ STRING(Number)

FOR count 1 TO Length(number string)

mychar $\leftarrow$ SUBSTRING(number string, count, 1)

myint $\leftarrow$ INTEGER(mychar)

Sum $\leftarrow$ Sum + myint

OUTPUT Sum

134

1+3+4 = 8